SolarMax

Testing the Solar Panel — AM33D Multimeter Only

Measure the panel tracking vs. not tracking. No resistors, no power supply.

Contents

1. The idea

You can evaluate the panel with **just the AM33D multimeter** by taking two readings on the bare panel:

- **Voc** — the open-circuit **voltage** (nothing connected but the meter).
- **Isc** — the short-circuit **current** (meter straight across the panel).

> **Note**
>
> **Why this shows the tracker working:** a panel's **short-circuit current (Isc) is proportional to how much sunlight lands on it** — so it rises when the panel faces the sun squarely and falls when it doesn't. Voltage (Voc) barely changes with aim. **So Isc is the number that proves tracking helps.**

The panel is a Renogy 50 W (RNG-50D-SS): in full sun expect **Voc ≈ 22 V** and **Isc up to ≈ 3 A**. Rough power: $P \approx 0.74 \times V_{oc} \times I_{sc}$.

2. Safety & setup (read first)

- Do this outside in **steady, strong sun** (clouds change the reading).

- **The panel does not connect to your power supply** — it makes its own power from light. Only the panel and the meter are involved.

Important

Shorting a solar panel is safe (unlike a battery) — it can only push its *Isc*, about 3 A. But two AM33D rules matter:

- For current, use the **10 A jack + 10 A dial position**. Do **not** use the `VΩmA` jack for this — it's fused at 250 mA and 3 A would blow that fuse.
- The datasheet limits the 10 A range to **15 seconds at a time**. Take the Isc reading quickly, then remove the probes.

3. AM33D quick settings

Black lead is always in **COM**. You'll use exactly two setups:

Reading	Dial position	Red lead jack	Typical display
Voc (voltage)	**200** in the DC-V block (V with =)	`VΩmA`	`21.8` (volts)
Isc (current)	**10A** in the DC-A block	`10A`	`2.85` (amps)

Note

Use the **DC** side (the "V" with a solid line above a dashed line, $V =$), *not* the AC "V∼". If the voltage shows a leading **"−"**, your leads are just reversed — no harm, the number is still correct. `OL` means over-range (shouldn't happen on these ranges here).

4. Measurement 1 — Open-circuit voltage (Voc)

1. Red lead in the `VΩmA` jack, black lead in `COM`.
2. Turn the dial to **200** in the **DC voltage** block.
3. Touch the red probe to the panel's + wire and black to −.
4. Read the voltage. In full sun it should be around **18–22 V**.

Tip

Press **HOLD** to freeze a reading so you can write it down. Voc changes only a little with aim — that's expected; the action is in the current (next).

5. Measurement 2 — Short-circuit current (Isc)

1. Move the red lead to the **10 A** jack (black stays in `COM`).
2. Turn the dial to **10A** in the **DC current** block.
3. Touch the probes across the panel's + and − wires — this briefly shorts the panel *through the meter*, which is fine.

4. Read the current **within a few seconds**, then lift the probes. Full sun, aimed at the sun: up to about **3 A**.

Important

Keep each Isc reading **under 15 seconds**. When you're done measuring current, **move the red lead back to the `VΩmA` jack** so you don't accidentally short something later while the meter is in high-current mode.

6. The experiment — tracking vs. not tracking

Do both readings twice, back-to-back, in the same sunlight:

1. **Tracking:** panel aimed **straight at the sun** (tracker running and pointed at it, or aim it by hand so its face is square to the sun). Record Voc, then Isc.
2. **Not tracking:** leave the panel at a **fixed, off-sun angle** (e.g. flat/horizontal, or rotated ~45° away). Record Voc, then Isc.
3. Repeat the pair at a few times of day (morning, noon, afternoon) — the gap is biggest when the sun is low.

Time	**Tracking**		**Not tracking**		**Isc gain**
	Voc (V)	Isc (A)	Voc (V)	Isc (A)	(%)
Morning					
Noon					
Afternoon					

7. What the numbers mean

- **Isc gain** $= \dfrac{I_{sc}(\text{tracking}) - I_{sc}(\text{not})}{I_{sc}(\text{not})} \times 100\%$. A positive gain is the tracker capturing more sunlight — that's your result.
- **Approx. power** for any reading: $P \approx 0.74 \times V_{oc} \times I_{sc}$. Compare the tracking vs. not-tracking power at each time.
- You should see **Voc nearly unchanged** but **Isc (and power) clearly higher when tracking** — especially morning and afternoon.

Note

For a fair comparison, take each tracking / not-tracking pair **quickly and in the same light**. If a cloud passes, redo that pair.

8. Troubleshooting

- **Current reads `0` or blows a fuse:** you used the `VΩmA` jack for current. Use the **10 A** jack and the **10A** dial position.

- **Voltage shows `OL`:** wrong (too-low) range — you're on DC **200**, which is correct for ~22 V; make sure it's DC, not AC.
- **Leading "–" sign:** leads are reversed. Fine — swap them or just note the value.
- **Readings jump around:** passing clouds or the panel is warming/shading. Wait for steady sun.
- **Meter turned off by itself:** auto power-off after 15 min — just turn it back on.

Meter details: `datasheets/ABMUAM33DRD-EN-v1.pdf` (AstroAI AM33D).